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# AI/ML for Nuclear Thermal Hydraulics

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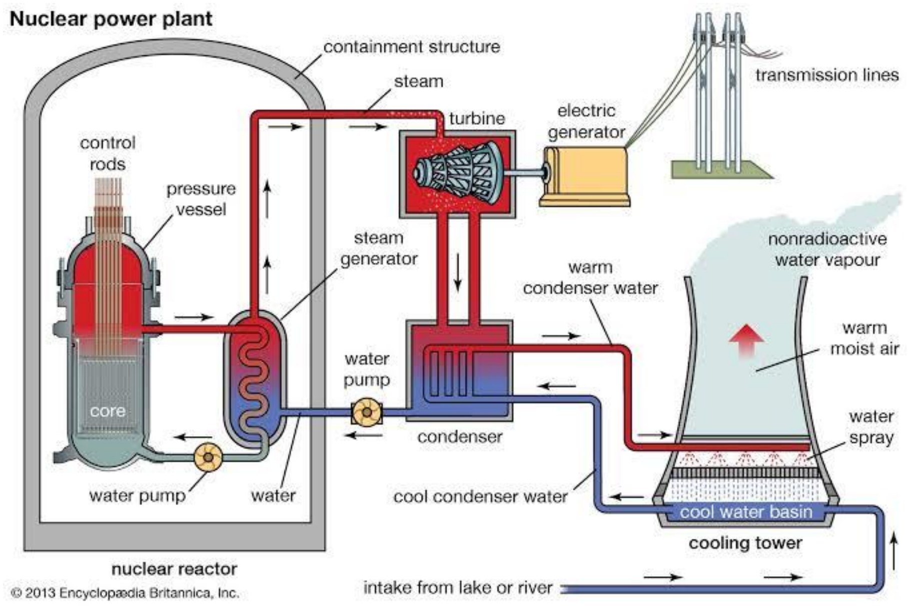
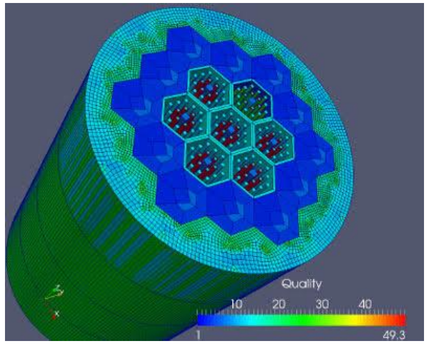
# Safety Requirements

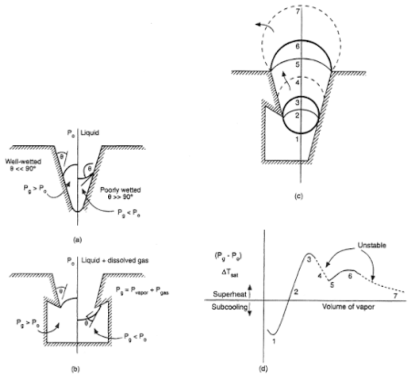
- All nuclear reactors rely on a working fluid (coolant) in order to transport energy from the core
- Unique to nuclear is
  - Its ability to operate at arbitrary power
  - Decay heat: significant heat (initially ~7% of operating generation) is still generated even after fission has ceased
- In all conceivable conditions, we must maintain a **margin to thermal limits** determined by the **structural integrity** of key components
- Conceivable conditions: determined by experience and available expert knowledge
- Inherently coupled to other physics (neutronics, structural)
- Inherently transient (& stiff)
- Inherently uncertain (aleatoric & epistemic)

# Thermal hydraulics challenges

- Nuclear thermal hydraulics is truly multiscale (in space and time)
  - System-scale: heat exchange throughout entire primary circuit, and secondary circuit
  - Core-scale: heat exchange from fuel pins to coolant in the core
  - Component-scale: flows through key components (plena, pressurizer, T-junctions, steam generators, valves & pumps)
  - Micro-scale: nucleate boiling, DNB, CHF

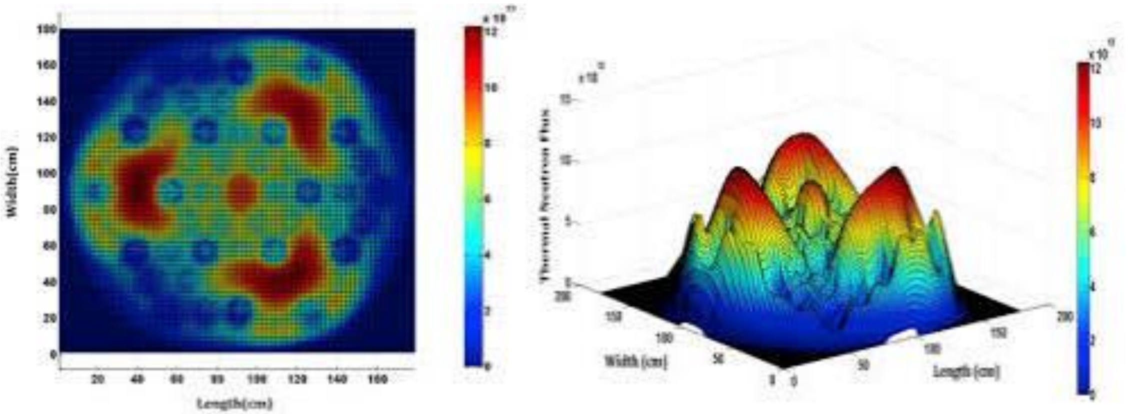


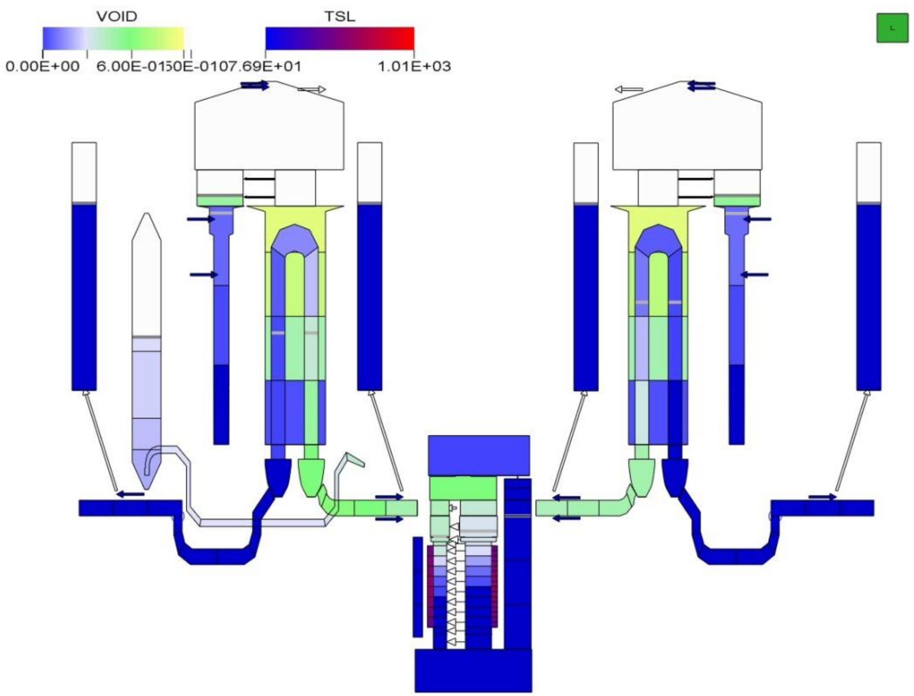
# Thermal hydraulics challenges

- NTH is also a multiphysics problem
  - Coolant properties affect the neutronics
  - Neutronics determines local heat generation
  - Local heat generation determines coolant properties
  - and so on…



# Simulation of TH systems: Systems codes

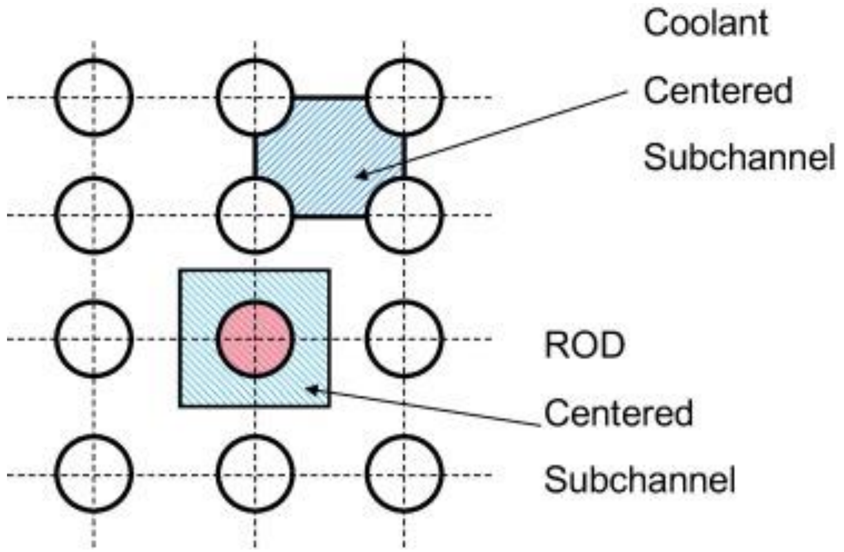
- System-scale: Systems codes
  - eg RELAP5-3D, ATHLET, CATHARE
- Simplified physics surrogate
  - Cross-section averaged channel flow
  - 1D ‘nodalization’
- Flow regime map; relates flow state to wall/interface correlations
- 0D models for pumps, valves & trips
- Rapid (~real time)

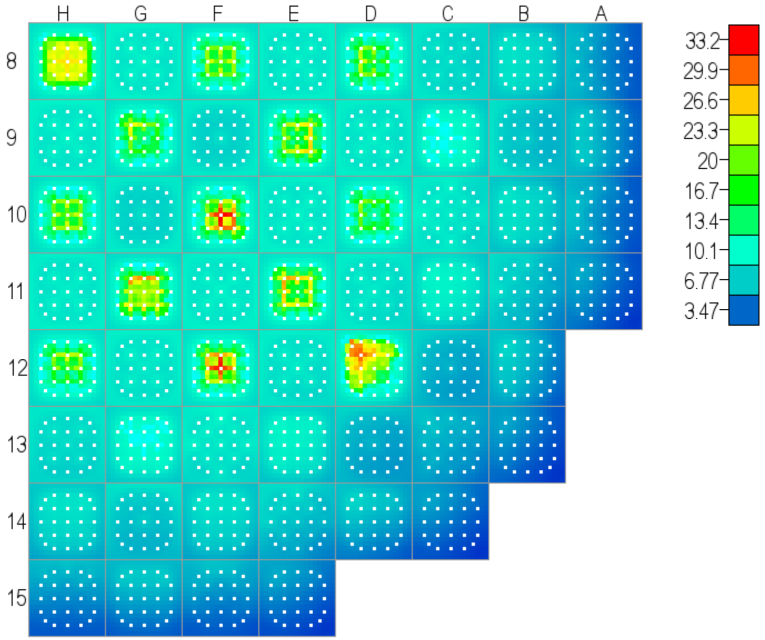


# Simulation of TH systems: Subchannel codes

- Core-scale: Subchannel codes
  - eg CTF, VIPRE, SUBCHANFLOW
- Simplified physics surrogate
  - Quasi 3D
  - Provides for lateral cross flow
- Flow regime map; relates flow state to wall/interface correlations

- Subgrid CFD

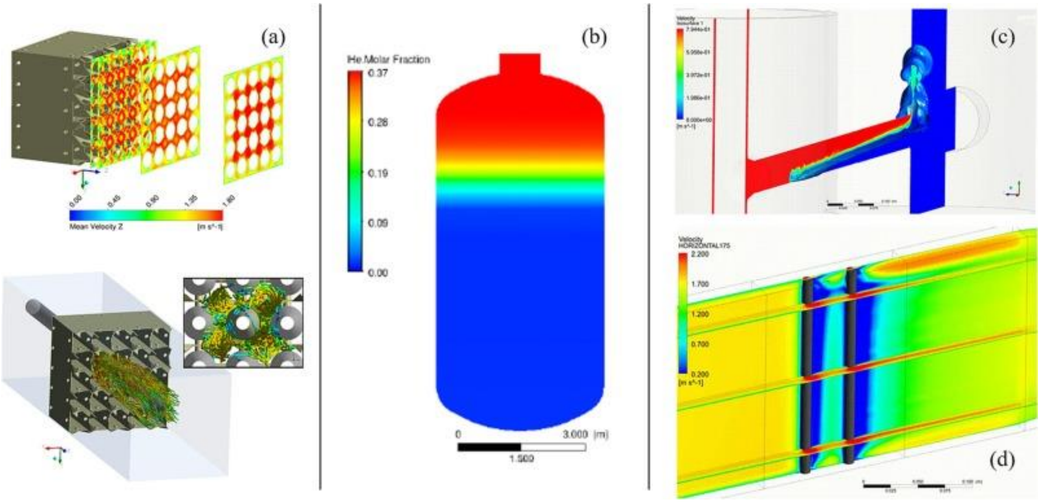


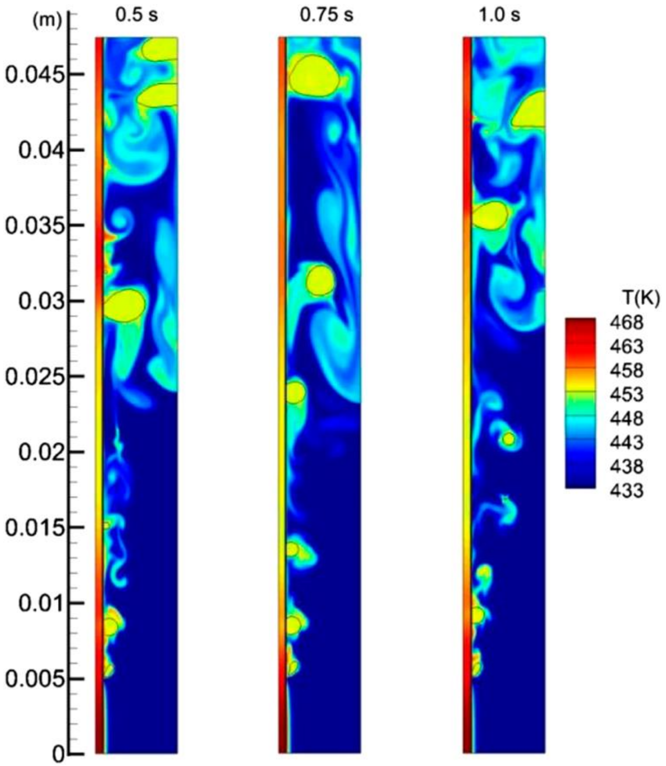




pin_avg_crud_thickness: Axial 266.112, exposure 2.46

# Simulation of TH systems: CFD

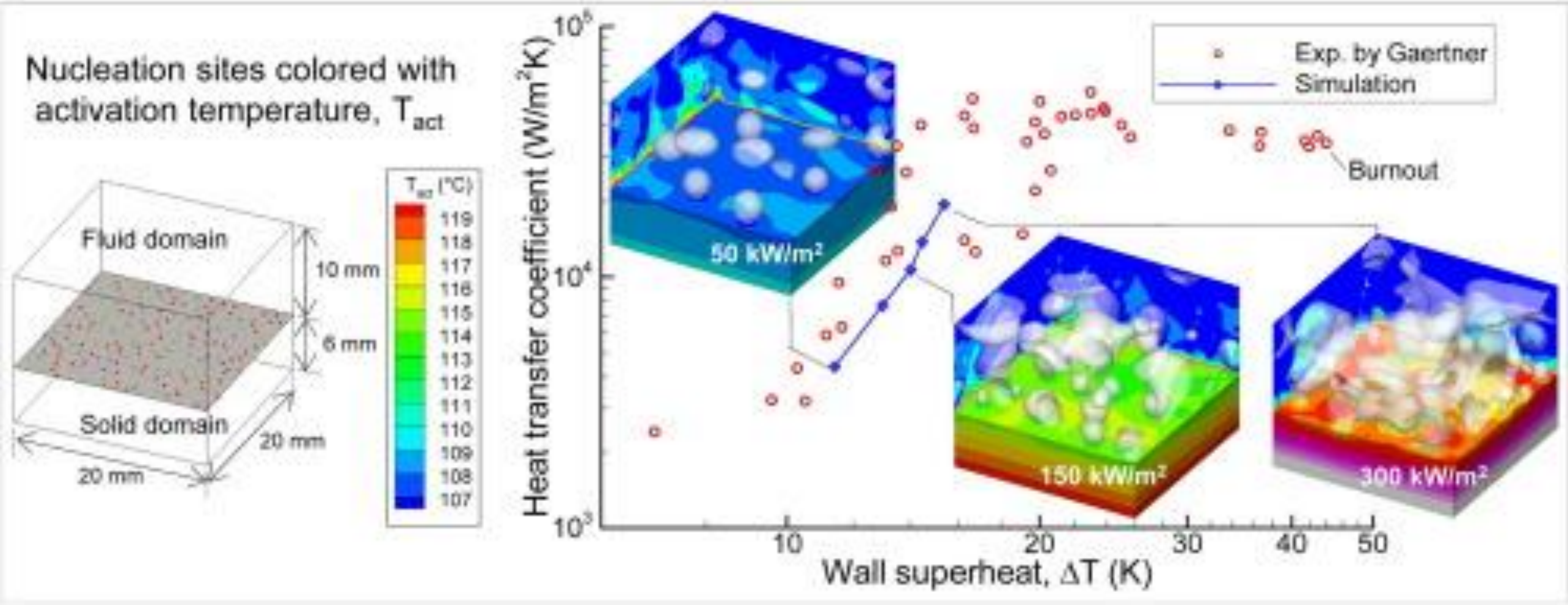
- Component-scale: Computational fluid dynamics (CFD)
- LES
- URANS
- RANS
- Multiphase
  - Eulerian-eulerian
  - Lagrangian
- Interface tracking



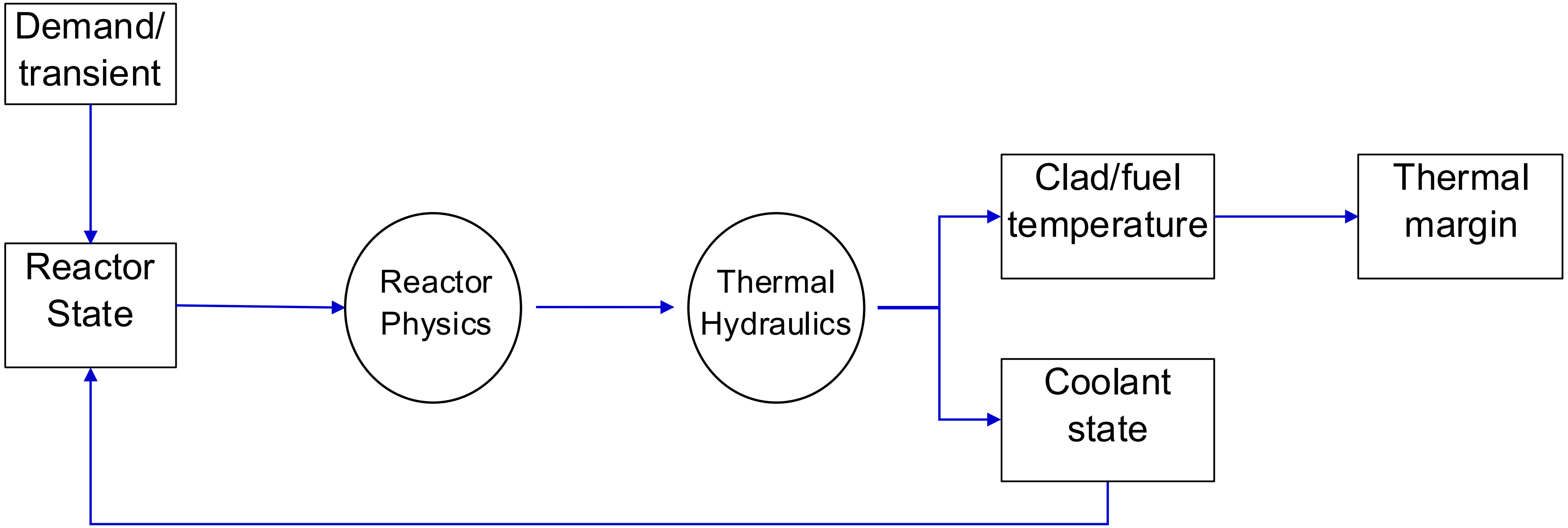
# Simulation of TH systems: Interface tracking

- Micro-scale: CFD
- Interface tracking
- Molecular dynamics



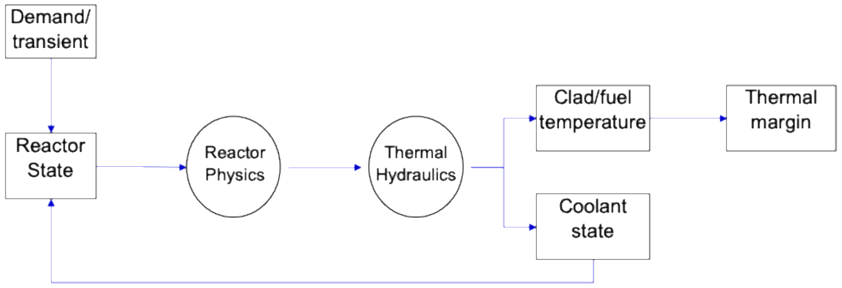
# Reactor analysis workflow

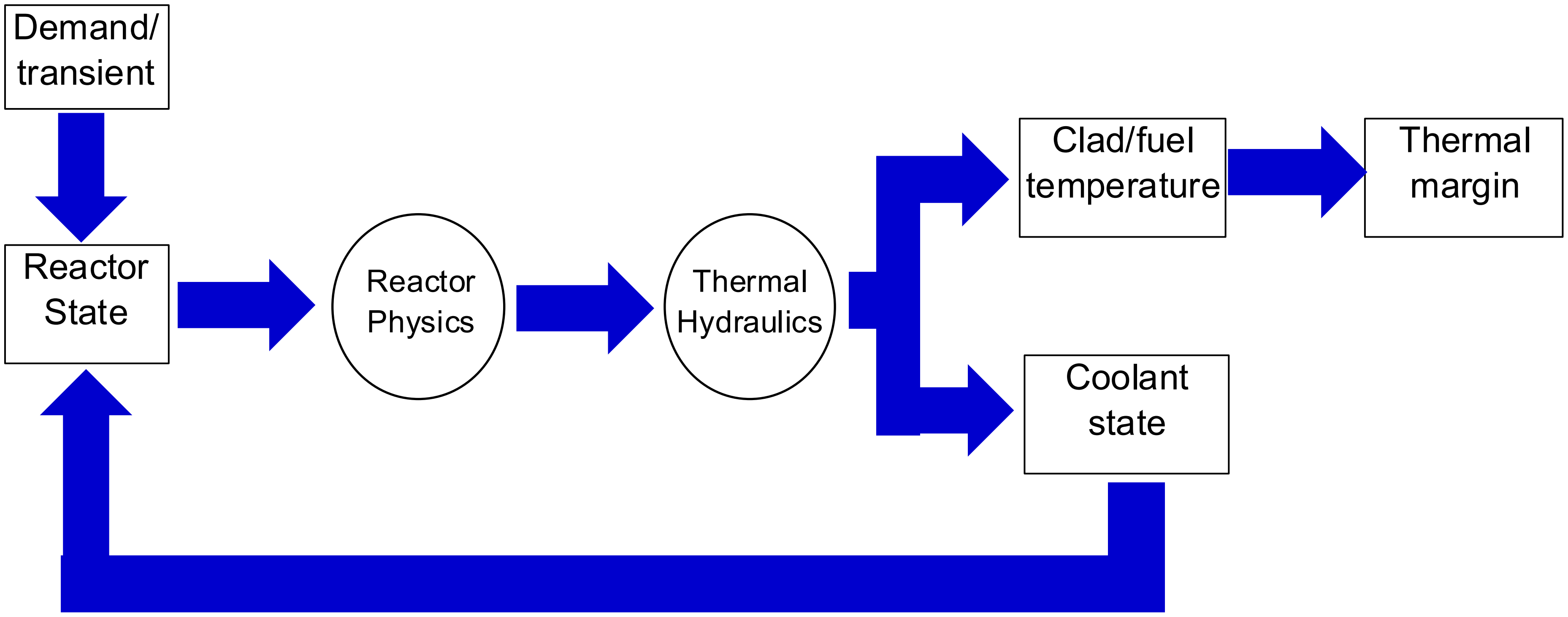
- Point-wise analysis



# Conservative workflow

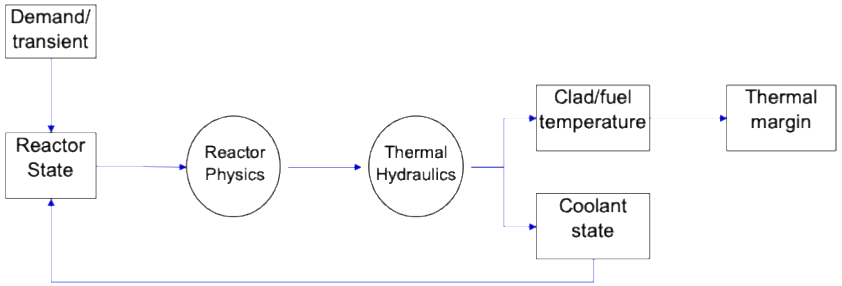
- Consider 'worst' cases (how?)



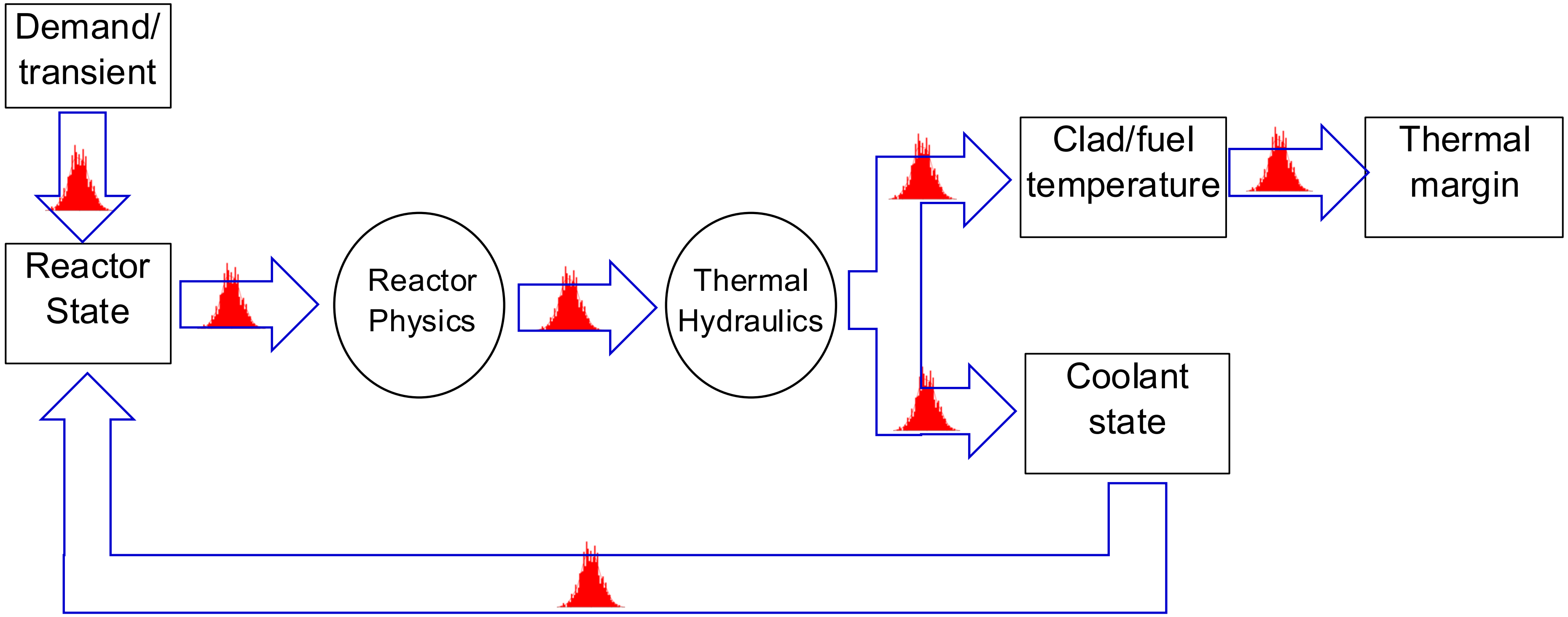




Likely to underpredict 'real' thermal margins, limiting performance

# Uncertainty aware workflow




- Consider uncertainties
  - epistemic and aleatoric

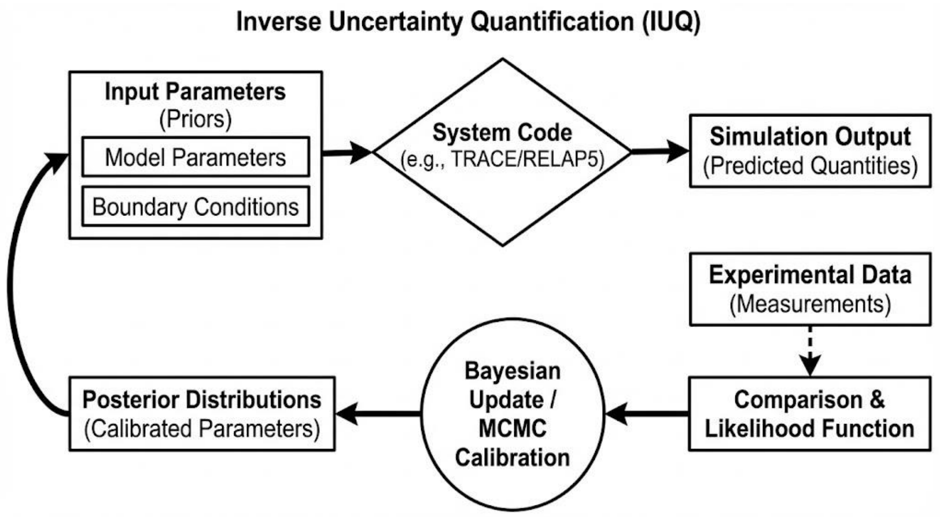



Better prediction of 'real' thermal margins, reducing conservatism

**Sampling requires many point-wise predictions**… hence need **'fast' models**

# Reactor analysis workflow

- Safety Analysis
  - Conservative/evaluation models
  - Best Estimate Plus Uncertainty (BEPU)
  - Inverse uncertainty quantification (IUQ)
- VVUQ
  - Verification
    - Establish that the code/method does what it is intended to do
  - Validation:
    - The model accurately (to some degree) reproduces experimental data
  - UQ
    - How accurate?
    - Establish error margins, sensitivities & output bounds

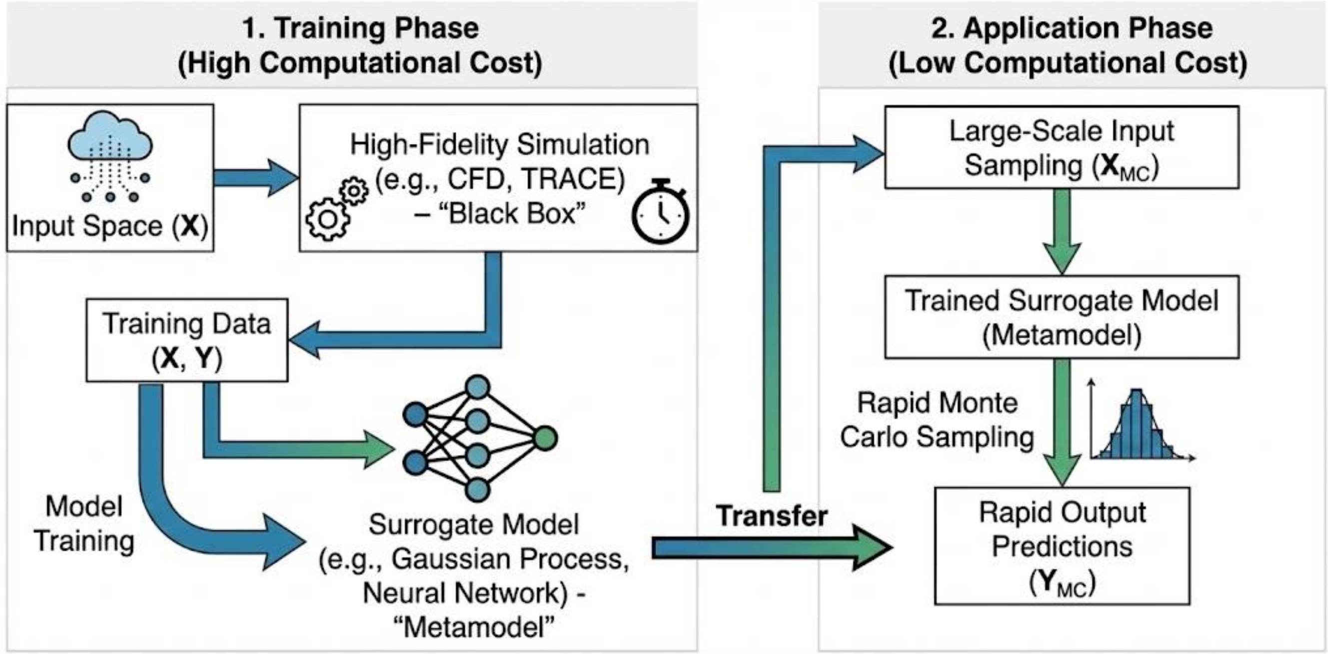


# Fast models

- MOR (Physics based)
  - Simplified physics (eg LES CFD, RANS CFD)
- Surrogates (Data-driven)
  - Look-up tables (LUT) (eg Groeneveld)
  - Correlation-based (eg Dittus-Boelter, etc)
  - Polynomial response surface
  - Polynomial chaos expansion
  - Proper orthogonal decomposition (intrusive & non-intrusive)
  - Kriging (Gaussian Process Regression) – note, provides UQ directly
  - **AI/ML**
- Hybrid MOR/surrogates
  - Systems/subchannel codes use simplified physics, LUTs & correlation-based surrogates

# AI/ML applications to NTH – Fast surrogates

- Data driven models
  - Random Forest, Gradient boost,…
  - Gaussian process regression
  - Neural networks
  - Many, many more
- Physics guided ML (PGML)
  - Physics informed neural networks (PINN)
    - Enforces conservation laws/PDEs as loss functions
  - Constrained architectures
    - Physical priors, kernel functions (eg in GPR)
  - Physics encoded NN (eg POD encoding)
- Generalizability
- Interpretability

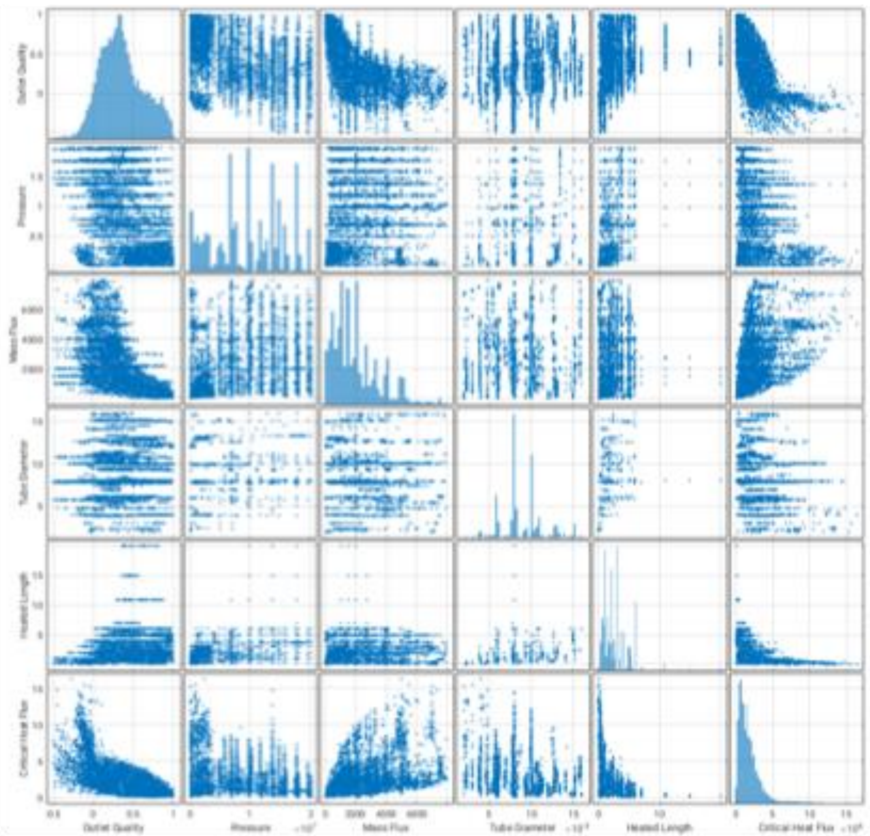


# AI/ML applications to NTH - Generalization

## Example: Critical Heat Flux

- Data: The Groeneveld Look-up table
  - A curated historical data-set (~30,000 points)
    - Inputs: Inlet quality, temperature, mass flowrate
    - Outputs: Critical heat flux
    - Uniform axial heating; circular pipe flow
- OECD/NEA AI/ML CHF Benchmark; Phase 1
  - ~50 submissions
  - Everything from decision trees, random forests, NN, GPR,…
  - Good 'in-data' results; great variability 'out-of data'
- OECD/NEA AI/ML CHF Benchmark; Phase 2
  - Generalizability: Non-uniform axial heating; bundle flows?
  - Uncertainty quantification?

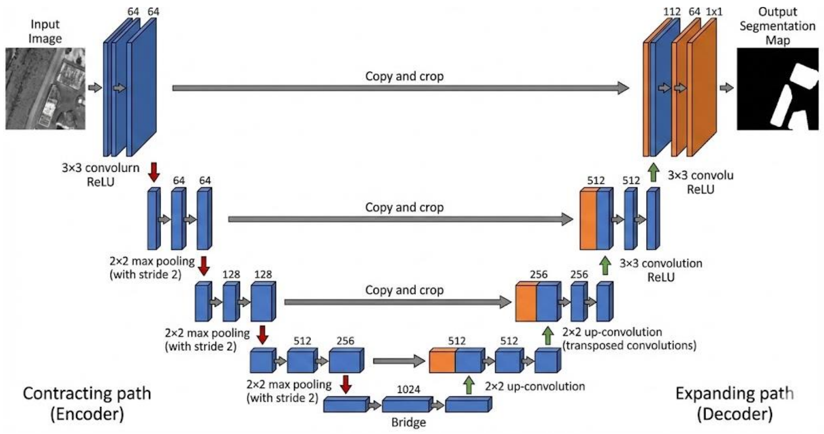
Figure 2.1. Scatter plot matrix of the NRC CHF database showing the relationship between pair of variables

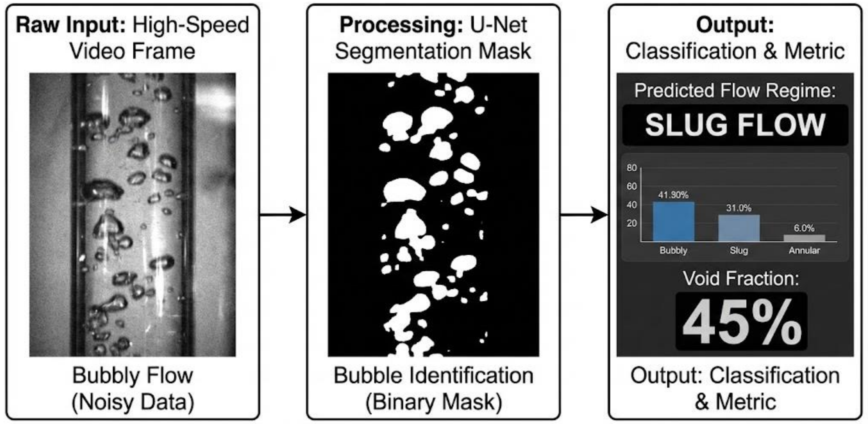


- Can AI/ML *learn* physics from the GLUT and generalize?

# What can AI/ML do for NTH? – Feature identification

- Feature identification



# Schedule

- Day 1:
  - NTH problem framing, supervised ML, and workflow of ML
  - Lab: Baselines, preprocessing pipelines, and first ML baseline model.
- Day 2:
  - Tree ensembles for engineering tabular data.
  - Lab: Tree ensembles with regime-wise diagnostics.
- Day 3:
  - Surrogate modelling and Gaussian Processes
  - Lab: GP surrogate and extrapolation stress test.
- Day 4:
  - Neural networks for tabular engineering data, limits, and lightweight physics-guided guardrails
  - Lab: basic feed-forward neural network, guardrails, and edge-case testing.
- Day 5:
  - Practical VVUQ, and safe optimisation for decision support.

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# Thank you